

# Chamod Shyamal Madusan

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## SUMMARY

I am a proactive recent graduate with a B.Sc. (Hons) in Electronic and Telecommunication Engineering from the University of Moratuwa. With strong programming skills and a solid foundation in mathematical thinking, my interests lie primarily in systems and high-performance computing, statistical learning, generative AI.

## EDUCATION

### • University of Moratuwa

Sri Lanka

B.Sc. (Hons) Electronic and Telecommunication Engineering

June 2021 - June 2025

–GPA: 3.76/4.00 ( First Class )

–Mathematics: A+

–Calculus: A+

–Methods of Mathematics: A

–Introduction to Engineering Optimization: A

–Probabilistic System Analysis: A-

–Pattern Recognition: A+

–Applied Statistics: A

–Deep Learning for Vision: A+

### • Richmond College , Galle

Sri Lanka

G.C.E Advanced Level Examination 2019 (Physical Science Stream)

Jan. 2006 - Dec. 2019

–Ranked: 50th in the country and District 5th with a z-score of 2.7754 (Top 0.28% )

### • Other Courses

–Trainee Full Stack Developer - University of Moratuwa (Open Learning Platform)

–Deep Learning Specialization - DeepLearning.AI (Coursera)

–Game Design and Development with Unity 2020 Specialization - Michigan State University (Coursera)

–Generative AI with Large Language Models - DeepLearning.AI , Amazon Web Services (Coursera)

–Deep Neural Networks with PyTorch - IBM (Coursera)

## INTERNSHIP EXPERIENCE

### • MAS Holdings Bodyline Pvt Ltd - Research Engineer (AI/ML)

2023 Dec - 2024 May

- Research Developed a computer vision model for error detection in labels , utilizing Raspberry Pi for model deployment, and created a local web app to run the error detection model on Raspberry Pi.
- Optimize model for embedded-level deployment and accelerate using Google Coral TPU.
- Developed PLC program for reconstructing and programming Automated Guided Vehicles (AGVs), emphasizing security and reliability.
- Other Skills: SolidWorks, Project Management, Electronics

## PROJECTS

### • Enfluent - AI powered language learning platform (enfluent.skillsurf.lk)🌐

2024-25

Developed a personalized language learning platform.

- Focuses on learning reading, writing, and speaking based on IELTS criteria.
- Provides individual AI grading for each criterion to evaluate progress.
- Created a 3D AI chatbot avatar provides real-time feedback on speech, including grammar, pronunciation, and confidence levels, enhancing learner engagement.
- Incorporated data from embedded systems (e.g., heart rate and emotional analysis) to adapt lesson plans and provide tailored feedback for individual learners.
- LLM consistency in writing tasks is modeled as the probabilistic method.
- Tools & technologies used: LLMs ,TTS and STT , Deepface , OpenAI , Python , ThreeJS , React , JavaScript , Rhubarb , Java

### • Ancient Language Character Translation 🐙

2025

Developed a Sinhala ancient handwritten character recognition system using deep learning and ensemble models.

- Designed a two-stage classification system for 88 ancient Sinhala characters and their historical eras using deep learning and traditional ML.
- Applied transfer learning with MobileNetV2 and fine-tuned dense layers for era classification across 37 datasets.
- Extracted deep features using VGG19, ResNet50, InceptionV3, and InceptionResNetV2, followed by ensemble classification.
- Enhanced model robustness through advanced data augmentation including shearing, brightness shift, and zoom transformations.
- Tools & technologies used: Python, TensorFlow, Keras, Scikit-learn, MobileNetV2, OpenCV, XGBoost

## • Brain Haemorrhage Detection 🧠

2023

Developed a brain hemorrhage detection and localization system to identify intracranial hemorrhages.

- Three models were analyzed, their performance metrics were defined, and ResNet150 was chosen for its effectiveness in identifying intracranial hemorrhages.
- Six windowing methods were utilized, and the sigmoid BSB method was identified as the most effective.
- Preprocessing data from a CSV file, comparing with the SOP Instance UID, and then opening the corresponding DICOM image and labels..
- Tools & technologies used: *Python , Tensorflow , Numpy , Pandas*

## • AI-powered Fitness Recommendation System 🧘

2024-25

Developed a microservices-based fitness platform providing personalized AI-driven recommendations.

- Built using a microservices architecture with asynchronous communication through RabbitMQ for scalable and decoupled system design.
- Implemented OAuth2 authentication for secure user access and protected personalized fitness data.
- Used Gemini AI model to analyze user data and generate fitness and wellness recommendations.
- Tools & technologies used: *Spring Boot, RabbitMQ, OAuth2, React, Gemini AI, Java, JavaScript, PostgreSQL, MongoDB*

## • Cyborg Fury - Third Person Shooting game Design & Develop 🎮

2024-25

Create an RPG shooting game with realistic 3D graphics using the Unreal game engine.

- Developed core mechanics like third-/first-person mode switching, weapon pickups, dodging, and tactical movement (ladder, elevator, cover).
- Includes intelligent enemy AI and customizable graphics for enhanced game-play
- Designed intelligent enemy AI with a player-buddy system for cooperative gameplay.
- Added vehicle control for spaceships and sci-fi vehicles to enhance exploration and combat.
- Implemented melee finishers, drones, turrets, power-ups, and sniper functionality for combat variety.
- Created a customizable UI with graphics settings and a fully functional main menu.
- Built with Unreal Engine 5.4 and Blueprints, the game offers immersive realistic graphics.

## • Open world Game Implementation Using Unity with Questionnaire Web Application🌐

2024

Start an exciting 3D RPG adventure in a open world , made for Unity WebGL platform.

- Beautifully optimized worlds and an engaging narrative.
- Includes an open-world adventure with realistic mechanics and quests like shooting, hunting, farming and stealing.
- Includes advanced weapons, vehicle control, a shop system, dynamic weather, and realistic gameplay.
- Develop a questionnaire website focused on promoting energy-saving habits and sustainable practices.
- Tools & technologies used: *Unity Engine , Character Creator 4 , Iclone 8 , C# , Java , JavaScript , React*

## TECHNICAL SKILLS AND INTERESTS

**Programming Languages:** Python , C++ , C# , JavaScript , Java , MATLAB

**Developer Tools:** HTML , CSS , Numpy, Pytorch , Tensorflow , React

**EDA, Simulation and CAD Tools:** Unreal Engine , Unity , Altium, Proteus, SolidWorks, Webots , Wokwi

**Languages:** : Sinhala (native) , English

## VOLUNTEERING AND LEADERSHIP EXPERIENCE

- **Department Facilitator** , EXMO23 The Flagship Technological Exhibition of University of Moratuwa 2023
- **Events Pillar** , ACIES seson 02 Gaming Competition organized by Mora Esports Club 2022
- **Committee Member** , Esports Club of University of Moratuwa 2021 - Present
- **Basket Ball Vice Captain** , Richmond College Galle 2017-2018
- **Treasurer** , Richmond College Science Society 2018

## ACADEMIC ACHIEVEMENTS

- **Mahapola Higher Education (Merit) Scholarship** 2020  
*For outstanding performance in GCE A/L Examination.*
- **The Most Deserving Student enter the University** 2020  
*Richmond College Galle , Sri Lanka*
- **Physics Olympiad - Silver Medal** 2019  
*Institute of Physics Sri Lanka*

## REFERENCES

### Dr. Ranga Rodrigo

*B.Sc. Eng. (Moratuwa), M.E.Sc. (Western, Canada), Ph.D. (Western, Canada), MIEEE*

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### Prof. Dileeka Dias

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